DO NOW — RECALL

Data Structures · Classes and Objects in C#

Name: _____ Class: _____ Date: _____

1. What is a method parameter?

2. What is a breakpoint?

3. How does a queue differ from stack?

4. Evaluate (A AND NOT B) OR C when A=1, B=0 and C=0. Show the intermediate Boolean values.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Named input to a method

2. Model answer

A debugger stopping point

3. Model answer

FIFO versus LIFO

4. Model answer

NOT B=1; A AND NOT B=1; then 1 OR C=1. Final output: 1.

By the end, you can…

• Define a class and instantiate an object

• Explain fields and methods

• Construct a simple record-like object

Vocabulary — know these terms

Recall the term — class

Recall the term — object

Recall the term — field

Recall the term — method

Recall the term — instance

NEW LEARNING

Read, annotate and explain

Separate blueprint and instance

```
Student pupil = new Student();
pupil.Name = "Asha";

class Student {
public string Name = "";
}
```

Class ≠ instance

A class defines the fields and behaviour available to each object. new Student() constructs an instance; a separate new creates another independent object.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

One class, two independent objects

Identify class, objects and members

Badge is the class (blueprint). amber and blue are references to two objects . Each has its own Points field and can run the Add method.

Trace the state change

new Badge() creates each object with Points = 0 . amber.Add(2) changes amber’s field from 0 to 2; blue remains at 0 because it is a different instance.

Predict, then check the output

The output is 2,0 . A class definition does not create an object. A method call on one object does not automatically change the fields of another.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

1. Different class, different data

What are first.Value and second.Value ?

Method / working:

2. Change the receiver

Continue from Example 1: second.Increase(); What values are stored now? Explain what second identifies.

Method / working:

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Apply independently
Write C# to define a class Book with public string Title. [3]

Question 2

Apply independently
Instantiate a Book object named novel. [2]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Apply independently
Explain why two objects of the same class may hold different field values. [2]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

True or false: defining a class automatically creates an object.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.